

Homework 7

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Physics 16

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Question 1. Purcell chapter 7, exercise 7.27 (Ring in a solenoid.) An infinite solenoid with radius b has n turns per unit length. The current varies in time according to $I(t) = I_0 \cos(\omega t)$ (with positive defined as shown in Purcell Fig. 7.36). A ring with radius $r < b$ and resistance R is centered on the solenoid's axis, with its plane perpendicular to the axis.

- (a) What is the induced current in the ring?
- (b) A given little piece of the ring will feel a magnetic force. For what values of t is this force maximum?
- (c) What is the effect of the force on the ring? That is, does the force cause the ring to translate, spin, flip over, stretch/shrink, etc.?

Solution. For part (a), the magnetic field inside the solenoid is $B = \mu_0 n I_0 \cos(\omega t)$. The flux through the ring of radius r is

$$\Phi_B = B \cdot \pi r^2 = \mu_0 n I_0 \pi r^2 \cos(\omega t).$$

By Faraday's law, the induced EMF is

$$\mathcal{E} = -\frac{d\Phi_B}{dt} = \mu_0 n I_0 \omega \pi r^2 \sin(\omega t).$$

making

$$I_{\text{ind}} = \frac{\mathcal{E}}{R} = \frac{\mu_0 n I_0 \omega \pi r^2}{R} \sin(\omega t).$$

For part (b), the force on a small piece of the ring is $dF \propto I_{\text{ind}} \cdot B$, since $d\vec{F} = I_{\text{ind}} d\vec{\ell} \times \vec{B}$. This product is proportional to

$$I_{\text{ind}} \cdot B \propto \sin(\omega t) \cos(\omega t) = \frac{1}{2} \sin(2\omega t).$$

This is maximized when $\sin(2\omega t) = \pm 1$, giving

$$t = \frac{\pi}{4\omega} + \frac{n\pi}{\omega}, \quad n \in \mathbb{Z}^+$$

For part (c), the magnetic field points along the solenoid axis ($\hat{z}$), and the induced current flows tangentially ($\hat{\phi}$). The force on each piece of the ring is

$$d\vec{F} = I_{\text{ind}} d\vec{\ell} \times \vec{B} \propto \hat{\phi} \times \hat{z} \propto \hat{r}.$$

The force on every piece of the ring points radially (either all inward or all outward, depending on the sign of $I_{\text{ind}} \cdot B$ at that moment). By symmetry, there is no net translational force, no net torque to spin or flip the ring. The force causes the ring to stretch or shrink radially. $\square$

Question 2. A metal crossbar of mass m slides without friction on two long parallel conducting rails a distance b apart. See Figure 7.35 in Purcell. A resistor R is connected across the rails at one end; compared with R , the resistance of the bar and rails is negligible. There is a uniform magnetic field perpendicular to the plane of the figure.

- (a) If the bar moves with velocity v toward the right, what is the current in the resistor? In what direction does it flow?
- (b) What is the magnetic force on the bar?
- (c) If the bar starts at time $t = 0$ with velocity v_0 , what is the speed at a later time?
- (d) The initial kinetic energy of the bar was $(1/2)mv_0^2$. Check that the energy dissipated by the resistor is the same.

Solution. For part (a), we know that the motional emf is

$$\mathcal{E} = Bbv.$$

Hence, the current in the resistor is

$$I = \frac{Bbv}{R},$$

and by Lenz's law it flows in the direction that opposes the increasing flux. For part (b), the current-carrying bar of length b sits in the magnetic field B . Since the current and field are perpendicular, the force is

$$F = BIb = B \cdot \frac{Bbv}{R} \cdot b = \frac{B^2b^2v}{R}.$$

By Lenz's law, this force opposes the motion (assume bar is moving $+x$ direction), so

$$\vec{F} = -\frac{B^2b^2v}{R} \hat{x}.$$

For part (c), applying Newton's second law,

$$\begin{aligned} m \frac{dv}{dt} &= -\frac{B^2b^2}{R} v \\ \frac{dv}{v} &= -\frac{B^2b^2}{mR} dt \\ [\ln v]_{v_0}^{v(t)} &= -\left[\frac{B^2b^2}{mR}\right]_0^t \implies \ln \frac{v(t)}{v_0} = -\frac{B^2b^2}{mR} t \implies \boxed{v(t) = v_0 e^{-B^2b^2t/(mR)}}. \end{aligned}$$

For part (d), we use $I = Bbv/R$ and part (c) to calculate power

$$P(t) = \frac{B^2b^2v(t)^2}{R} = \frac{B^2b^2v_0^2}{R} e^{-2B^2b^2t/(mR)}.$$

Then,

$$E = \int_0^\infty P(t) dt = \frac{B^2b^2v_0^2}{R} \int_0^\infty e^{-2B^2b^2t/(mR)} dt = \frac{B^2b^2v_0^2}{R} \cdot \frac{mR}{2B^2b^2} = \boxed{\frac{1}{2}mv_0^2}.$$

□

Question 3. Consider a “tank circuit” (an inductor and a capacitor connected in series) with $L = 5 \times 10^{-2}$ H and $C = 5 \times 10^{-6}$ F. At $t = 0$ there is 1.44 mJ of energy stored in the inductor and 1.44 mJ of energy stored in the capacitor, and the capacitor is discharging.

- (a) How much charge is on the capacitor at $t = 0$ s?
- (b) How much current is in the inductor at $t = 0$ s?
- (c) How much time will elapse (in seconds) until the first instance when the stored energy is entirely magnetic?
- (d) How much time will elapse (in seconds) until the first instance when the stored energy is entirely electric?

Solution. The LC circuit obeys the differential equation

$$L \frac{dI}{dt} + \frac{Q}{C} = 0 \implies L \frac{d^2Q}{dt^2} + \frac{Q}{C} = 0 \implies \frac{d^2Q}{dt^2} + \omega_0^2 Q = 0,$$

where $\omega_0 = 1/\sqrt{LC}$. The general solution is

$$Q(t) = Q_0 \cos(\omega_0 t + \phi).$$

Then, current is

$$I(t) = -\frac{dQ}{dt} = Q_0 \omega_0 \sin(\omega_0 t + \phi).$$

First, compute ω_0 :

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{(5 \times 10^{-2} \text{ H})(5 \times 10^{-6} \text{ F})}} = 2000 \text{ rad/s}.$$

For part (a), the energy stored in the capacitor is $U_C = Q^2/(2C)$. At $t = 0$,

$$\frac{Q(0)^2}{2C} = 1.44 \times 10^{-3} \text{ J} \implies Q(0) = \sqrt{2C \cdot 1.44 \times 10^{-3} \text{ J}} = \sqrt{2(5 \times 10^{-6} \text{ F})(1.44 \times 10^{-3} \text{ J})}.$$

$$Q(0) = \sqrt{1.44 \times 10^{-8} \text{ C}^2} = \boxed{1.2 \times 10^{-4} \text{ C} = 120 \mu\text{C}}.$$

For part (b), the energy stored in the inductor is $U_L = LI^2/2$. At $t = 0$,

$$\frac{LI(0)^2}{2} = 1.44 \times 10^{-3} \text{ J} \implies I(0) = \sqrt{\frac{2 \cdot 1.44 \times 10^{-3} \text{ J}}{L}} = \sqrt{\frac{2.88 \times 10^{-3} \text{ J}}{5 \times 10^{-2} \text{ H}}}.$$

$$I(0) = \sqrt{5.76 \times 10^{-2} \text{ A}^2} = \boxed{0.24 \text{ A} = 240 \text{ mA}}.$$

Now apply the boundary conditions to find Q_0 and ϕ . At $t = 0$,

$$Q(0) = Q_0 \cos \phi = 1.2 \times 10^{-4} \text{ C}, \tag{1}$$

$$I(0) = Q_0 \omega_0 \sin \phi = 0.24 \text{ A}. \tag{2}$$

Dividing (2) by (1) gives

$$\tan \phi = \frac{I(0)}{\omega_0 Q(0)} = \frac{0.24 \text{ A}}{(2000 \text{ rad/s})(1.2 \times 10^{-4} \text{ C})} = \frac{0.24 \text{ A}}{0.24 \text{ A}} = 1 \implies \phi = \frac{\pi}{4}.$$

Now, we find the amplitude Q_0 :

$$Q_0 = \frac{Q(0)}{\cos(\pi/4)} = \frac{1.2 \times 10^{-4} \text{ C}}{1/\sqrt{2}} = 1.2\sqrt{2} \times 10^{-4} \text{ C}.$$

So the full solutions are

$$Q(t) = 1.2\sqrt{2} \times 10^{-4} \text{ C} \cdot \cos(2000 \text{ rad/s} \cdot t + \pi/4),$$

$$I(t) = 0.24\sqrt{2} \text{ A} \cdot \sin(2000 \text{ rad/s} \cdot t + \pi/4).$$

For part (c), the energy is entirely magnetic when $Q = 0$,

$$\cos(2000 \text{ rad/s} \cdot t + \pi/4) = 0 \implies 2000 \text{ rad/s} \cdot t + \pi/4 = \pi/2$$

$$2000 \text{ rad/s} \cdot t = \pi/4 \implies t = \frac{\pi}{8000} \text{ s}$$

$$\boxed{t = \frac{\pi}{8000} \text{ s} \approx 3.93 \times 10^{-4} \text{ s.}}$$

For part (d), the energy is entirely electric when $I = 0$,

$$\sin(2000 \text{ rad/s} \cdot t + \pi/4) = 0 \implies 2000 \text{ rad/s} \cdot t + \pi/4 = \pi$$

$$2000 \text{ rad/s} \cdot t = \frac{3\pi}{4} \implies t = \frac{3\pi}{8000} \text{ s}$$

$$\boxed{t = \frac{3\pi}{8000} \text{ s} \approx 1.18 \times 10^{-3} \text{ s.}}$$

□